

Workplace Application Course (WAP 228)

The course you are taking under the course titled “Spring Semester Workplace Application” consists of three phases. In the first phase, you will participate in various training sessions. These trainings are mandatory; please upload the certificates for the trainings you complete in the first phase to the form at <https://forms.office.com/r/Kj0CdW5Y4Q> . You can view the relevant courses under the “Workplace Application Phase 1” section.

You will have the opportunity to apply the skills you acquired during Phase 1 in Phase 2. Two projects have been assigned for you in this phase; you will select and complete one of these two projects. You can find the details of the projects in the Workplace Training Phase 2 section. You will submit the completed project via GitHub in Phase 3. You can find the key points to note for each phase below under the headings Workplace Application Phase – 1, Workplace Application Phase – 2, and Workplace Application Phase – 3.

Important – Trainings and Project Topic Selection

All courses (tracks) in Phase 1 are mandatory for every student without exception; you must complete them all. Your choice of programming language only comes into play during project development in Phase 2. You will develop your project using a single technology stack; you will integrate your chosen language with SQL and GitHub within the same project. You will choose one of the following two combinations:

- Python + SQL + GitHub (all three integrated into the same project)
- Java + SQL + GitHub (all three integrated into the same project)

Students who successfully complete this Workplace Application course can apply for positions such as data analyst, machine learning engineer, web developer, data engineer, business intelligence analyst, software developer, and game developer at companies. They can apply for junior positions and engineering internship programs at top-tier companies.

Workplace Application Phase – 1

All of the tracks listed below are mandatory training modules that every student must complete in order to successfully pass the Workplace Applications course. Please complete all of them on DataCamp and upload the certificates for the tracks you’ve completed to the appropriate section in the [form](#).

Required Tracks (For All Students)

1. Data Literacy
2. Python Data Fundamentals
3. Java Fundamentals
4. SQL Data Fundamentals
5. GitHub Foundations

Note: Your choice of programming language (Python or Java) does not affect these courses; all five tracks are mandatory for everyone. Your language choice applies only during Phase 2, when you are developing your project.

Workplace Application Phase – 2

In this phase, you will first select your technology stack: Python + SQL + GitHub or Java + SQL + GitHub. The programming language, SQL, and GitHub you choose will be used together within the same project. You will then select one of the two projects below and develop it using your chosen stack. Paste the GitHub link of your completed project to the designated section on the [form](#). Your use of both your programming language, SQL, and GitHub will be evaluated together in the project.

1st Project — Database Application Project: Library Management System

Description: You will develop a library management system that connects to an SQL database using the programming language of your choice (Python or Java). The system must support the creation, listing, updating, and deletion (CRUD) of book, member, and loan records; all data must be stored in an SQL database. You must manage the entire development process in a GitHub repository using meaningful commits.

Ana Kullanılması Gerekenler:

- SQL scripts that define the database schema (CREATE TABLE, relationships, and sample data insertion/INSERT).
- Source code that establishes a database connection and performs CRUD operations in your chosen language (sqlite3/SQLAlchemy for Python and JDBC for Java are recommended).
- A command-line interface (CLI) or a simple graphical user interface (GUI) for user interaction.
- Test cases demonstrating the system's functionality.
- The project's GitHub repository: a consistent commit history, descriptive commit messages, and a README file containing the system design and usage instructions.

2nd Project — Data Analysis and Reporting Project: Analysis of Olist E-Commerce Sales Data

Description: In this project, you will use the “Brazilian E-Commerce Public Dataset by Olist” dataset on Kaggle (<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>). The dataset is not a single table; it consists of multiple tables (CSV files) linked by keys, such as orders, order items, customers, products, sellers, payments, and reviews. You will import these tables into an SQL database, establish relationships between them, and perform meaningful sales analyses using SQL queries (specifically JOIN, GROUP BY, aggregate functions, and filtering). You will then process, visualize, and report on the query results in your chosen language. You will manage the entire project in a GitHub repository.

Sample analysis questions you can answer: changes in total orders and revenue by month; top-selling product categories; sales distribution by state/city; usage rates of payment methods; the relationship between customer review scores and delivery times.

Essential Points:

- SQL scripts that load Olist tables into the database; SQL queries containing JOINS written for schema definition and analysis to establish relationships (foreign keys) between tables.
- Source code that connects to the database in your chosen language, processes the results, and visualizes them (using pandas + Matplotlib/Seaborn for Python; data retrieval via JDBC and graph/output generation for Java).
- A report summarizing the findings and insights derived from the dataset
- The project's GitHub repository: README, source code, SQL scripts, and analysis outputs.

General Guidelines For Your Project

- Documentation: Add comments to the code and include a README file that provides a general overview of the project, instructions on how to run the code, and the required dependencies
- Version Control: You are required to use Git and GitHub for your project. Make meaningful and frequent commits to track changes and regularly back up your work; include a link to your repository in your documentation.
- Database: You are expected to demonstrate your SQL skills in your project. Proper table design, relationships, and queries will be evaluated.

Workplace Application Phase – 3

In this phase, you will submit the project you developed in Phase 2 via GitHub. Create a GitHub repository for your project, upload (push) all source code, SQL scripts, and documentation to this repository, and share the repository link in the designated section on the [form](#). Ensure that your repository is public or accessible for evaluation.

In your repository's README file, include the following: how you developed the project and the process you followed; the challenging parts of the project; what you learned by the end of the project; and the roles played by the programming language, SQL, and GitHub in the project. The README should also include a general summary of the project, instructions on how to run the code, and the necessary dependencies.

A well-organized and meaningful commit history is part of the evaluation, as it demonstrates the development process of your work.

In this phase, you are required to write a report for your final project. **You will include your report in the form mentioned above and are required to submit all documents by 23.59, June 23rd, 2026 to the form. The template for the project will be shared with you within a week.**

Good Luck

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